

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

PHASE 24

3-STAGE HANGAR SCRUB BOUNDARIES & CORRIDOR BIO-GEL CASSETTES

Air blast — bio-gel panels — chemical/thermal shower
500mm cassettes every 50m — collect daily, grow ~3 days, read



SUITS: AUTOMATED HIGH-INTENSITY UV/THERMAL CYCLES — NO HUMANS IN THE LOOP

500mm BIO-GEL CASSETTES — HEAD-HEIGHT EVERY 50m — COLLECT DAILY, GROW ~3 DAYS, READ

GP-IM-024

Revision v1.0 — 20 August 2026

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VETTED: PASS — ORDINARY ENGINEERING, HONEST ASSAY

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LICENSE: CERN OPEN HARDWARE LICENCE v2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 24

PHASE 24: 3-STAGE HANGAR SCRUB BOUNDARIES & CORRIDOR BIO-GEL CASSETTES — STATUS:

CURRENT. The ship's clean-line discipline: every hangar junction runs a three-stage scrub boundary — high-pressure air blast, physical contact with vertical bio-gel static trapping panels, chemical/thermal shower — with suits cycling automated high-intensity UV/thermal, no humans in the loop. And in the corridors: the ribbon array — 500mm bio-gel cassettes at head height every 50 metres, collected daily, growth-cabinet read over ~3 days, then scanned, UV-sterilized, re-melted and re-poured in an infinite material-free loop. Vetted: PASS — the colour-change reading was the auditor's own AI-interpretation, withdrawn by him; the assay is the author's, seated.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-024, v1.0 — 20 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the scrub law as seated (contents line and both design bodies, verbatim, with the author's clarification riding) — §2 why it works (graded) — §3 the system on the ship — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Establishes automated atmospheric scrub lines at all hangar junctions, backed by reactive corridor bio-gel cassettes that visually discolor upon detection of airborne microscopic contaminants or pressure variations.' [CLARIFIED BY THE AUTHOR 12 AUGUST 2026: the colour-change reading was never the design — see the assay below; the line stays preserved verbatim per doctrine.]

THE SEATED DESIGN BODY — THE BOUNDARY (verbatim): 'Hangar Scrub Boundary: Ground crews pass through a 3-stage boundary: Stage 1 (High-pressure air blast); Stage 2 (Physical contact with vertical bio-gel static trapping panels); Stage 3 (Chemical/thermal shower). Environmental suits undergo automated

high-intensity non-human UV/thermal cycles.'

THE SEATED DESIGN BODY — THE RIBBON ARRAY (verbatim, clarification bracket included): 'Air-Monitoring Ribbon Array: Standardized 500mm bio-gel cassettes mounted at human head-height every 50m along all main corridors are rotated manually every 24 hours. Exposed gel is scanned, sterilized via high-intensity Phase 10 UV lasers, re-melted, and re-poured in an infinite, material-free closed loop.

[CLARIFIED BY THE AUTHOR 12 AUGUST 2026 (vet batch 11): the assay is collect daily → replace the cassette → growth cabinet for ~3 days → observe growth. The colour-change reading was never the design — the external auditor owned it as his own AI-interpretation and withdrew it. Test spec owed: assay sensitivity and target organisms.]'

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE BOUNDARY PHYSICS: three-stage air isolation/scrubbing is ordinary engineering — the vet's own words, PASS. Air blast strips the loose; bio-gel static panels trap the clinging; chemical/thermal shower kills the rest; the suits get their own automated non-human UV/thermal cycle so the boundary never trusts a person to decon a person.

THE ASSAY PHYSICS, CORRECTED AND SEATED: the cassette is a COLLECTOR, not a chameleon. The reading is biological: collect daily → replace the cassette → growth cabinet ~3 days → observe growth. What grew is the instrument. This is culture-based bio-monitoring — slow, honest, sensitive — the same discipline as any cleanroom settle plate, flown on a starship.

THE LOOP ECONOMY: exposed gel is scanned, sterilized by high-intensity Phase 10 UV lasers, re-melted, re-poured — an infinite, material-free closed loop. The consumable is never consumed; only the information is taken.

THE HONEST LIMITS, SEATED: a growth assay answers in days, not seconds — it is a trending instrument, not an alarm. The alarm layer is other hardware. And the owed test spec is the author's own bracket: assay sensitivity and target organisms — what the cabinet is meant to reveal (§9).

§3 — THE SYSTEM ON THE SHIP

- THE BOUNDARY: every hangar junction, three stages in order — air blast, gel panels, shower — ground crews pass through; the line is procedural law, not architecture you can walk around.
- THE SUITS: automated high-intensity UV/thermal cycles, non-human — suits are deconned by machine, never by their wearers.
- THE RIBBON: 500mm cassettes, head-height, every 50 metres of main corridor, manual 24-hour rotation — the corridor itself is the sampling grid.
- THE LOOP: scan → UV-sterilize (Phase 10 lasers) → re-melt → re-pour — the cassette stock is eternal;

the record is what changes.

INTEGRATION NOTE (audit, flagged): the cassettes and the boundary plumbing are pod-and-corridor furniture — placement rides the Phase 19/32 pod-wall standard and the corridor grid (GP-IM-019 v1.0); UV laser fittings ride Phase 10's doctrine. Quantities and staffing ride §9.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE AVIAN EXCEPTION (seated law): 'Poultry is restricted entirely to sterile eggs, incubated exclusively within Phase 24 closed biocontainment systems.' — the scrub doctrine owns the ship's highest-risk biology.
- THE TERRESTRIAL ROLLOUT : 'Deploys scaled variants of the Phase 24 bio-gel detector ribbon panels within commercial shopping complexes, high-rise towers, and hospital wings.' — the corridor ribbon is an Earth product too (the Corporation's subsidiary line).
- THE BIO-CONTAINMENT FAMILY : the Phase 12 inward-extension chain runs its own steam/chemical pre-wash on retrieved cores — the scrub boundary is the hangar-side sibling of the bio-siphon discipline (GP-IM-012).
- THE CLUSTER RECORD (seated): Phase 24 rode two combined headers — the 12-cluster (its content's home) and the 6/8/24 header (the naming batches' doing); both remnants stand with their split notations (sitting 498).

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Build the boundary hardware: air-blast vestibule, vertical bio-gel static trapping panels, chemical/thermal shower — three stages, in order, at every hangar junction.
- STEP 2 — Automate the suit line: high-intensity UV/thermal suit cycles, non-human; interlock the boundary so no stage is skippable.
- STEP 3 — Build the cassette: the standardized 500mm bio-gel unit — gel formula, frame, head-height mount; stock the 50-metre grid.
- STEP 4 — Commission the rotation: manual 24-hour collection, replacement stock, the growth-cabinet line (~3 days), the observation record — the assay IS the instrument.
- STEP 5 — Commission the loop: scan, Phase 10 UV sterilize, re-melt, re-pour; verify sterile reset across cycles (§9).

- STEP 6 — Write the test spec (§9): target organisms, assay sensitivity, action thresholds — his to seat; then the array is operational.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 12 — the agar bio-siphons (GP-IM-012): the bio-containment family's retrieval side; the boundary's doctrine sibling.
- PHASE 10 — the Foamies UV plant: the high-intensity UV lasers that sterilize the gel loop (§1, §3).
- PHASE 19 — the blast gates (GP-IM-019 v1.0): the corridor grid and pod-wall standard the ribbon mounts to.
- PHASE 25 — the seeding pods (GP-IM-025): the fungi line — biological material crosses the boundary law like everything else.
- PHASE 26 — the snowglobe (GP-IM-026): the closed-loop agriculture the boundary protects; the cluster mate.
- PHASE 6 — Hull & Structure (GP-IM-006 v1.4): hangar junction architecture the boundaries seat in.

§8 — DO-NOT-BUILD

- NEVER read the cassette as a colour-change alarm — the author's clarification is seated law: collector + growth cabinet; the reading is biological (§2).
- NEVER let a human decon a suit — the suit cycle is automated and non-human (§1).
- NEVER treat the assay as real-time — it trends in days; alarms are other hardware (§2).
- NEVER break the loop with consumables — scan, sterilize, re-melt, re-pour; material-free means material-free (§1).
- NO skipped stages at the boundary — three, in order, interlocked (§6).

§9 — OWED

THE BENCH, OWED: THE AUTHOR'S OWN TEST SPEC — assay sensitivity and target organisms for the growth cabinet (his bracket, seated); action thresholds per organism class; sterile-reset verification across re-pour cycles; gel formula spec; boundary air-blast and shower cycle figures; cassette quantities per corridor-kilometre and the rotation staffing; Stage 2 panel capture efficiency. All owed items ride the bench until spoken.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-024, v1.0 — CURRENT, seated law, master v2.1 (numerical print order, post-split). Built 20 August 2026.

PROVENANCE — THE STANDING ORDER, VERBATIM (seated in sitting 499): '16 and then bed for me, after we finish that you can do 17 to 27 and i will read all 10 first thing to see if we can get through them faster in blocks'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart. Next version triggers: the author's test spec, the §9 benches, or his word.

END OF CHAPTER 24